

Ian Lewis

Software Engineer

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SUMMARY

Full-stack software engineer experienced across JavaScript, TypeScript, Python, Ruby, React, FastAPI, Node.js, AWS, Docker, PostgreSQL, and observability tooling. Recent work includes a self-hosted observability platform with cardinality control, a webhook inspection service, and a RAG application using vector search and the OpenAI API. Previous career in educational leadership adds strong communication, mentoring and project management skills.

TECHNICAL EXPERIENCE

Trickl, *Co-Creator, Software Engineer*

03/2026 – 06/2026

Trickl 🌐 is an open-source, scalable metrics observability platform, with built-in cardinality control measures for small to medium-sized development teams.

- Implemented a telemetry collection layer utilizing **Vector**, routing data to both a **VictoriaMetrics** backend for real-time querying capabilities and AWS **S3** for long-term storage.
- Designed and built an **OpenTelemetry** cardinality management system that identifies label's impact on total series count and surfaces ranked reduction recommendations, applied live via hot-reload with **zero service downtime**.
- Provisioned a production ready AWS environment using **CDK** (TypeScript), including **EC2**-backed **ECS**, **autoscaling groups**, persistent **EBS** volumes, **VPC networking**, security groups, **RDS** and **S3**.
- Ensured production readiness by incorporating **Cloud Map** for deployed service-to-service discovery, **WAF-based API key verification** of payloads at the load balancer layer, **Cognito OIDC**-guarded **Grafana** access control, per-container **CloudWatch** logging, and managed secrets lifecycle via **AWS Secrets Manager**.
- Designed a **React**-based visualization layer for **out-of-the-box simplicity** making Trickl easier to adopt.
- Integrated an **OpenAI**-backed **agentic workflow** using a 7-class JSON classification schema and a 3-round tool-use loop to generate user-facing context.
- Built a **Dockerized** local development environment and **CLI** to improve setup and testing consistency across local and **cloud** environments.
- Wrote a technical case study 🌐 documenting Trickl's problem domain, tradeoffs and key engineering decisions in order to **communicate complex architectural rationale clearly**.

Independent Projects & Client-Style Applications, *Software Engineer*

09/2023 – 03/2026

- The Dugout 🌐 (**Python, FastAPI, React, Vite, PostgreSQL, Pinecone, OpenAI API, Vercel, Render**). Ingested and chunked source material, generated embeddings for vector storage, orchestrated semantic search using the user query to retrieve and generate a response utilizing the OpenAI API and a token friendly model.
- RequestStore (**Node.js, Express, React, TypeScript, MongoDB**). Self-hosted RequestBin-style tool for capturing and inspecting incoming webhooks and HTTP requests. Utilized a custom AWS VPC for deployment making use of public & private subnets, AZs, internet and NAT gateways and Nginx as a reverse proxy.
- LineupCFB 🌐 (**React, JavaScript, Supabase, Edge Functions, Render, Vercel**). Live ESPN score ingestion, complex scoring logic and week advancement with historical snapshots. Deployed and actively maintained.

SKILLS

AI Engineering: Prompt Engineering, OpenAI API, RAG, Multi-Agent Orchestration, Mastra, Evals, Claude Code, Codex

Languages: Python, JavaScript, TypeScript, Ruby, SQL, HTML/CSS

Frontend: React, Vite, Tailwind CSS, Next.js

Backend: Node.js, Express, FastAPI, REST APIs, Zod, JWT, WebSockets

Cloud & Infrastructure: AWS, CDK, ECS, Docker, Linux, Unix, CI/CD, Git, Cloud Networking, Infrastructure as Code

Databases & Storage: PostgreSQL, MongoDB, Redis, S3, Supabase

Observability: Prometheus, OpenTelemetry, Grafana, VictoriaMetrics, Mimir, Honeycomb

Testing & Developer Tools: Jest, VS Code, CLI Tooling, Vitest, React Testing Library, Playwright, Postman

ADDITIONAL EXPERIENCE

Educator & Administrator, *UK & US School Systems*

08/2013 – 09/2023 | UK, US

- Managed budgets, mentored early-stage teachers, designed and delivered curriculum and led staff development initiatives designed to improve instructional quality and student outcomes.

EDUCATION

MBA, *Northeastern State University*

09/2021 – 12/2022

Graduated with Honors.

BS in Education, *Kingston University*

09/2010 – 07/2013 | London, England

Graduated with Honors.